Imprint Field Dynamics: Coherence–Entanglement Continuum and the Reinterpretation of the Double-Slit Experime

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Abstract
The Imprint Field Hypothesis reinterprets the quantum measurement process as a coupling between particle, field,

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1. Introduction
The double-slit experiment demonstrates the interplay between particle discreteness and wave interference. The Im

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2. The Imprint Field Hypothesis
The quantum system is defined as a triad: Particle (P), Field (F), and Imprint (I):

$$R = P + F + I$$

The field stores and redistributes phase information, analogous to ripples in water encoding a droplet’s motion. Whe

Field potential:

$$\Psi(r,t) = A(r,t)\, e^{i\phi(r,t)}$$

A encodes coherence amplitude; ϕ represents imprint phase.

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3. Coherence–Entanglement Continuum (CEC)
CEC describes the smooth transition between coherence and entanglement. The coherence parameter:

$$\gamma = \langle \Psi_P | \Psi_F \rangle / (\|\Psi_P\|\ \|\Psi_F\|), \quad 0 \le \gamma \le 1$$

$\gamma=1$ corresponds to perfect coherence; $\gamma=0$ to full entanglement.
Coherence energy:

$$U_c = \blacksquare\omega\blacksquare\ (1 - \gamma^2)$$

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4. Reinterpretation of the Double-Slit Experiment
Under the Imprint Field model, a particle generates a coherent imprint through both slits. The screen records interfe

$$I(x) = I\blacksquare\ [1 + \gamma \cos(\Delta\phi(x))]$$

Collapse occurs when $\gamma \to 0$.

Figure 1: Imprint Field Triad schematic.
Figure 2: Coherence–Entanglement Continuum diagram.

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5. Experimental and Technological Applications
Quantum Computing:
Coherence–Entanglement Binary (CEB) logic defines states as:

|0> = Coherent
|1> = Entangled
|2> = Intermediate

Triadic qutrits allow deterministic computation between coherence and entanglement.

Quantum Sensing: